SEMESTER/YEAR : III / II YEAR
COURSE CODE : 25MBAB001
TITLE OF THE COURSE : DATA MANAGEMENT SYSTEMS
L: T: P: C : 3: 0: 2: 4

Overview

Today 's organizations recognize that managing data is central to their success. They recognize data has value and they want to leverage that value. Data is the fundamental raw material for any business which enable sensing the internal and external parameters that could have bearing on business performance. Those parameters provide the basis for action when analyzed for insights. The insights are relevant to understand the current state of the business as well as the likely future state. The derived insights provide the basis for all business decisions which precede action. Having the required insights in time, enable the organizations to be responsive and adapt. Harnessing required data while ensuring its dependability organized storage to enable access on demand, data security and access control is the subject of Data Management Systems. As all aspects of business have dependency on data, every business executive has a definite role in contributing to the success of Data Management Systems. Business organizations are adapting to become digital enterprises of the future. Understanding Data Management Systems facilitates active participation in transformation process to Digital Enterprise.

Course Objectives

1. To enable students to appreciate data requirements for business.
2. To enable students to understand Data and Data Management Systems in relation to any business domain by a systematic process based on Goal Setting and Performance Management as the factors that lead to success.
3. To enable understanding of data sources, data capture & storage.
4. To enable students to understand data models, Master Data, MDM, Schema, Relational and No-SQL databases, Data Warehouse and Data Lake.
5. Enabling understanding of data organisation from speed of response, and information security.

Course Outcomes

By the conclusion of this course, the student should be able to

1. Exhibit the fair understanding of concepts, theories, models and principles of business data management course, including data requirements, business entities, their relationships, and the business value of data.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of business data management in real-time problem solving by identifying and harnessing various data sources such as ERP systems, big data, databases, data

warehouses, NoSQL databases, ETL systems, IoT, and data streams.

3. Demonstrate the ability of analysing real-time problems using the skills acquired from concepts, theories, models and principles of business data management by evaluating master data, organizational data dictionaries, and one-truth databases.
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of business data management for decision-making related to data centres, including in-house, co-located, and cloud computing (private, public, hybrid).
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of business data management to formulate data security requirements and policies.

Approach to Learning	• Lectures • Readings • Active student participation and classroom exercises • Case Analysis collaboratively with students' involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Finance practices by way of: • Mini projects, • Submission of assignments • Group assignments • Written Exam

Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	**Data Fundamentals** i. Data, Types of Data, information, derived insights from Analysis, Meta Data. ii. Data Structures. iii. Decision Making in Management - Strategic, Tactical and Operational decisions. iv. Decision Support with Management Information Systems. v. Data Sources & Defining Data Need - Internal Data. Transaction data structured and unstructured data. - External Data. PESTEL, Competition, Porters 5 Forces. - Physical Records, Flat Files, Spreadsheet, Databases, social media. - Data Models - Retails marketing as a case study to exemplify definition of data requirements. - Information Security.	8

Uni t II	**Data Base Systems, Relational Databases (RDBMS) in Particular.** • Entity Relations and ER Diagram. • Database Schema. • Data base organization, Primary key, and foreign Key. • Stored Procedures and Referential Integrity. • Data Base Replication. • Introduction to SQL for query and reporting with examples. • Popular RDBMS with relative merits.	8
Uni t III	**Using SQL for extracting information from RDBMS.** • Introduction to Entity and Entity Relations. • Attributes. • Entity – Primary and foreign keys. • SQL commands. • Using MS-SQL Express to work on creating, modifying, Querying, reporting.	8
Uni t IV	**Data Warehouse, Data Lakes and their Applications.** • Data Warehouse and Data Mart. • Datawarehouse Schema, star, snowflake, star- • Operational Data Store. • ETL • OLAP. • Limitations of Data Warehouse. • Data Lake, concepts, Business Purpose. • Well known Data Warehouse and Data Lake Products.	8
Uni t V	**Big Data, Unstructured Databases aka No SQL Databases, Data Mining.** • Big Data, concepts and uses. • Introduction and the need for unstructured data bases. • organization of No SQL Databases. • Top Rated No-SQL Databases. • Extracting Information from No-SQL Databases. • Introduction to Data Mining, business purpose served, Data structures for Data Mining.	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	3	-	-	-	2
CO 2	3	-	-	-	2
CO 3	3	-	-	-	2
CO 4	3	-	-	-	2
CO 5	3	-	-	-	2

Action Based Components

- Defining data requirements for Finance, Marketing & Sales, HR, SCM and CRM.
- Use SQL Query to extract information from MS-SQL Express. Students will be installing MS-SQL express, load data and use SQL to query and extract data.
- Design a Mini Data warehouse.
- Work on No-SQL using Mongo-SQL or any other No-SQL Databases depending on free license availability.
- Mini workshop on information security needs, access control.

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	2	3	4	5	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X	X			
6	CBT	10	X	X	X			

7	Semester End Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Primary: Recommend students to Invest in this book as it provides an excellent perspective about Information Management and its various applications with real world case studies. Management Information System Auth: C. Laudon Kenneth and P. Laudon Jane Pub: Pearson
2. Secondary (Optional) Master Data Management and Data Governance Auth: Alex Berson and Larry Dubov Pub: McGraw Hill.

Reference books

1. DAMA-DMBOK: Data Management Body of Knowledge Paperback – Illustrated, 4 July 2017 Publisher: Technics Publications LLC.
2. E-Business and E-Commerce Management – Author, Dave Chaffey, Publisher – Pearson Education.
3. Management Information Systems – Authors, James A. O'Brien, George M. Marakas, Ramesh Behl – McGraw Hill Education (India)
4. Data Warehousing, Data Mining & OLAP – Authors Alex Berson, Stephen J. Smith – Tata McGraw-Hill
5. Big Data: Big Data, Black Book: Covers Hadoop 2, MapReduce, Hive, YARN, Pig, R and Data Visualization by DT Editorial Services.
6. Data Lake: Data Engineering with Apache Spark, Delta Lake, and Lakehouse Auth Manoj Kukreja and Danil Zburivsky Pub: Packt
7. Principles of Information Security, Authors, Michael E. Whitman, Herbert J Mattord – Publisher, Cengage Learning.
8. The Fourth Industrial Revolution – Author, Klaus Schwab, Founder and Executive Chairman, World Economic Forum – Publisher, Penguin Random House.

Suggested Courses

1. IBM Data Engineering Professional Certificate | Coursera: https://www.coursera.org/professional-certificates/ibm-data-engineer

Online References

1. https://docs.oracle.com/database/121/TDPDW/tdpdw_arch.htm#TDPDW380
2. https://docs.informatica.com/data-integration/data-services/10-2/application-service-guide/content-management-service/reference-data-warehouse.html
3. https://www.bigdataframework.org/big-data-architecture/#:~:text=%20The%20NIST%20Big%20Data%20Reference%20Architecture%20,can%20be%20used%20by%20the%20Big...%20More%20
4. https://www.geeksforgeeks.org/data-architecture-design-and-data-management/
5. https://pubs.opengroup.org/onlinepubs/9199929899/toc.pdf

SEMESTER/YEAR : III SEM / II YEAR
COURSE CODE : 25MBAB002
TITLE OF THE COURSE : APPLIED ANALYTICS
L: T: P: C : 3: 0: 2: 4

Overview

The Applied Analytics course is designed not only to explain what each model does or functions but also explores how businesses use them, whether it is to gather insights, solve problems, or predict outcomes.

Course Objectives

1. Apply quantitative modelling and data analysis techniques to the solution of real-world business problems, communicate findings, and effectively present results.
2. Demonstrate knowledge of statistical data analysis techniques utilized in business decision making.
3. Apply principles of Data Science to the analysis of business problems.
4. Use data mining software to solve real-world problems.
5. Employ cutting edge tools and technologies to analyse (Big) Data.

Course Outcomes

By the conclusion of this course, the student should be able to

1. Exhibit the fair understanding of concepts, theories, models and principles of business analytics course, including how to collect and prepare data for analysis.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of business analytics in real-time problem solving by using analytical tools to gather meaningful business insights.
3. Demonstrate the ability of analyzing real-time problems using the skills acquired from concepts, theories, models and principles of business analytics through data-driven decision-making to support informed business decisions.
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of business analytics to examine whether analytical results confirm business hypotheses.
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of business analytics, including evaluating ways to apply machine learning techniques to solve business problems.

Approach to Learning	● Lectures ● Readings ● Active student participation and classroom exercises ● Case Analysis collaboratively with students 'involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Finance practices by way of: ● Mini projects, ● Submission of assignments ● Group assignments ● Written Exam

Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	**Spreadsheet Modelling** i) Concepts and Best Practices ii) Relative & Absolute Addresses, Range Names, Auditing Formulas iii) SUM, AVERAGE, PRODUCT, COUNT, COUNTA, COUNTBLANK, COUNTIF, SUMIF, AVERAGEIF, COUNTIFS, SUMIFS, AVERAGEIFS, SUMPRODUCT, IF, VLOOKUP, RAND, RANDBETWEEN iv) Pivot Tables, Data Tables, Goal Seek	8
Unit II	**Marketing** i) Predicting customer churn ii) Customer segmentation iii) Sales forecasting	8
Unit III	**Human Resource** i) Predictive HR analytics in recruitment and selection ii) Predictive HR analytics in turnover and separation iii) Predictive HR analytics in learning and development	8
Unit IV	**Finance** i) Predicting stock market returns ii) Detecting fraudulent transactions iii) Predicting bank loan defaults	8

Unit V	**Supply Chain** i) Monte Carlo simulation modeling (Harry ‘s Auto Tire) ii) Transportation problem (Executive Furniture Corporation) iii) Waiting line problem (Three Rivers Shipping Company)	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	3	-	-	-	2
CO 2	3	-	-	2	2
CO 3	3	-	-	2	2
CO 4	3	-	-	2	2
CO 5	3	-	-	2	2

Action Based Component

1. Advanced Excel Functions and Formulas using Microsoft Excel 2019
2. Supervised Machine Learning XLMiner
3. Unsupervised Machine Learning XLMiner
4. Deterministic Optimisation using Microsoft Excel Solver
5. Monte Carlo Simulation using Microsoft Excel 2019

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	2	3	4	5	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X	X			
6	CBT	10	X	X	X			
7	Semester End	40	X	X	X	X	X	

	Examination							

Recommended Resources

Textbook

1. Spreadsheet Modeling and Decision Analysis: A Practical Introduction to Business Analytics Author: Cliff Ragsdale, Edition: 8^{th}, Year: 2018, ISBN: 9789353502225, Publisher: Cengage India
2. XLMiner & Solver Platform for Education (add-ins to Microsoft Excel). All students should compulsorily have Microsoft Excel 2019 on their laptops

Reference Book

1. Shmueli, G., P. C. Bruce, and N. R. Patel, Data Mining for Business Analytics: Concepts, Techniques, and Applications with XLMiner, New Jersey, John Wiley & Sons, 2016.

Suggested Courses

1. https://www.coursera.org/lecture/predictive-modeling-analytics/7-how-to-build-a-model-using-xlminer-fM9fY
2. https://www.coursera.org/lecture/data-analytics-business-capstone/building-logistic-regression-models-using-xlminer-vl92D
3. https://www.coursera.org/lecture/predictive-modeling-analytics/7-building-neural-networks-using-xlminer-3eQXt
4. https://www.coursera.org/lecture/business-analytics-decision-making/5-cluster-analysis-with-xlminer-E53Hq

SEMESTER/YEAR : III SEM / II YEAR
COURSE CODE : 25MBAB003
TITLE OF THE COURSE : DATA VISUALIZATION FOR DECISION MAKING
L: T: P: C : 3: 0: 2: 4

Overview

In this course, students will learn the basics of visualization principles. How to convert data into powerful visualizations leading to actionable insights? Tableau is a Business Intelligence tool for visually analysing the data. Users can create and distribute an interactive and shareable dashboard, which depict the trends, variations, and density of the data in the form of graphs and charts. Tableau can connect to files, relational and Big Data sources to acquire and process data. The software allows data blending and real-time collaboration, which makes it very unique. It is used by businesses, academic researchers, and many government organizations for visual data analysis. It is also positioned as a leader Business Intelligence and Analytics Platform in Gartner Magic Quadrant.

Pre Requisites

Tableau Public Software

Course Objectives

With this course, students will learn the fundamentals of data visualization

1. Enable students with tools and techniques to communicate analytical insights
2. Empower students with understanding of appropriate visualization for the task in hand
3. Empower students with data blending options across data sources and bring them to life using visualizations
4. Enable them to create impactful dashboards for the problem in hand
5. Enable students to do storytelling for non-technical audience using data visualization

Course Outcomes

By the conclusion of this course, the student should be able to

1. Exhibit the fair understanding of concepts, theories, models and principles of data visualization and Tableau course, including interpreting analytical insights through appropriate visualizations.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of data visualization and Tableau in real-time problem solving by choosing appropriate visualizations for specific tasks.
3. Demonstrate the ability of analysing real-time problems using the skills acquired from concepts, theories, models and principles of data visualization and Tableau by examining data sources and bringing them to life through visualizations.
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of data visualization and

Tableau for selecting impactful dashboards for given problems.

5. exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of data visualization and Tableau, including developing analytical insights using Tableau story.

Approach to Learning	● Lectures ● Readings ● Active student participation and classroom exercises ● Case Analysis collaboratively with students' involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Finance practices by way of: ● Mini projects, ● Submission of assignments ● Group assignments ● Written Exam

Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	**Data Visualization Principles** Univariate Categorical Data, Univariate Numerical Data, 1 Categorical Data & 1 Numerical Data, Bivariate Categorical Data, Bivariate Numerical Data	8
Unit II	**Introduction to Tableau** Installing Tableau Public, Getting Started with Tableau, Saving Tableau Public, Navigation Design Flow, File Types, Data Types, Show Me, Terminology	8
Unit III	**Data Sources** Data Sources, Fields Operation: Adding, Combining, Searching, Reordering, Editing Metadata, Data Joining, Data Blending, connecting to Google Sheets Connecting to PDFs	8
Unit IV	**Tableau Features** Drill Down, Swapping Dimensions, Calculations, Sort & Filters, Tableau Parameter, Action URL, Actions Filter	8
Unit V	**Storytelling with Tableau** Tableau Charts: Bar, Line, Pie, Crosstab, Scatterplot, Bubble, Boxplot, Histogram, Tree Map Dashboards, Story, Formatting, Forecasting, Trend Lines, Storytelling Guidelines	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	3	-	-	3	-
CO 2	3	-	-	-	2
CO 3	3	-	-	-	2
CO 4	3	-	-	-	2
CO 5	3	-	2	3	2

Action Based Components

- Work with multiple datasets and data types
- Create tableau charts on multiple datasets
- Create dashboards on multiple datasets
- Create story on multiple datasets

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	**2**	**3**	**4**	**5**	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X	X			
6	CBT	10	X	X	X			

7	Semester End Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Communicating Data with Tableau (O 'Reilly Publishers), Ben Jones

Reference books

1. Storytelling with Data (Wiley), Cole Nussbaumer Knaflic

Suggested Courses

1. Data Visualization with Tableau UCDavis, University of California https://www.coursera.org/specializations/data-visualization?

Online References

1. https://www.tutorialspoint.com/tableau/index.htm
2. https://public.tableau.com/en-us/s/resources
3. https://www.tableau.com/learn/articles/free-public-data-sets

SEMESTER/YEAR : III SEM / II YEAR
COURSE CODE : 25MBAB004
TITLE OF THE COURSE : PREDICTIVE ANALYTICS USING R
L: T: P: C : 3: 0: 2: 4

Overview

In this course, students will learn the basics of machine learning using R programming.
They will get a perspective on:

- Where Machine Learning fits into Analytics landscape
- Types of Machine Learning algorithms
- Data Science process from data collection till model evaluation

Course will be hands-on using R and R Studio and mini projects will be implemented for each algorithm type.

Course Objectives

1. Learn R Data Structures
2. Learn Exploratory Data Analysis and its significance
3. Implement Supervised Machine Learning algorithms using R in a business application
4. Implement Unsupervised Machine Learning algorithms using R in a business application
5. Implement Time Series using R

Course Outcomes

By the conclusion of this course, the student should be able to

1. Exhibit the fair understanding of concepts, theories, models and principles of machine learning course.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of machine learning in real-time problem-solving by performing EDA and machine learning processes using R.
3. Demonstrate the ability of analyzing real-time problems using the skills acquired from concepts, theories, models and principles of machine learning by evaluating model performance with appropriate measures.
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of machine learning for real-time problem-solving and model appraisal.
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of machine learning, including choosing appropriate ML algorithms for different situations.

Approach to Learning	● Lectures ● Readings ● Active student participation and classroom exercises ● Case Analysis collaboratively with students' involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Finance practices by way of ● Mini projects, ● Submission of assignments ● Group assignments ● Written Exam

Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	**Introduction to ML** • Types of Analytics: Descriptive, Diagnostic, Predictive, Prescriptive • Statistics Basics • Types of Data: Structured, Semi Structured, Unstructured • Structured Data using RDBMS. Table, Columns, Rows • What is ML; ML Use Cases; Algorithm Types: Supervised, Unsupervised; Supervised, Reinforced; Regression Vs. Classification; • Managing and Understanding Data: Vectors, Factors, Lists, Dataframes, Matrixes and Arrays, Importing and Saving Data from CSV files • Exploring and Understanding Data: Exploring the structure of Data, Exploring Numeric Variables, Exploring Categorical Variables, Exploring Relationships between Variables	8

Unit II	**Supervised Machine Learning** ● KNN Algorithm: How does it work? ● Example - Diagnosing breast cancer with KNN ● Decision Trees: How does it work? Gini Index, Chi Square to choose decision node, Parameters to control overfitting ● Example – identifying risky bank loans ● Neural Networks: How do they work? ● Example - Modelling the strength of concrete with ANNs ● Multiple Linear Regressions ● Example – predicting medical expenses using linear regression	8
Unit III	**Unsupervised Machine Learning** ● Market Basket using Apriori algorithm, Support, Confidence, Lift ● Example – identifying frequently purchased groceries with association rules ● K Means algorithm Example - Finding teen market segments using k-means clustering	8
Unit IV	**Evaluating Model Performance** ● Measuring performance for classification ● Working with classification prediction data ● Using Confusion Matrix to measure performance ● Measures other than accuracy: kappa, sensitivity, specificity, precision and recall, F ratio, ROC ● Estimating future performance: holdout, cross validation, bootstrap sampling	8
Unit V	**Time Series Introduction** ● Data Types and formats ● Components and Objectives ● Reading, plotting and decomposing time series data in R ● Forecasts using exponential smoothing in R ● ARIMA models in R	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	3	-	-	2	2
CO 2	3	-	-	-	2
CO 3	3	-	-	-	2
CO 4	3	-	-	1	2
CO 5	3	-	-	1	2

Action Based Component

1. Exploratory Data Analysis using R, Orange
2. Supervised ML (KNN, Decision Tree, Neural Network) using R, Orange
3. Unsupervised ML (Clustering, Market Basket Analysis) using R, Orange
4. Model Evaluation using R, Orange
5. Time Series Using Excel, Dataiku

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	**2**	**3**	**4**	**5**	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X	X			
6	CBT	10	X	X	X			

7	Semester En d Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Machine Learning using R (Packt Publishing), Brett Lantz 2013
2. A little book of R for time series, Avril Coghlan

Reference books

1. Data Analytics with R, Dr Bharti Motwani, Wiley

Suggested Courses

1. EDA https://moderndive.netlify.app/1-getting-started.html
2. https://supervised-ml-course.netlify.app/ (4 Case Studies by Julia Silge, Data Scientist from RStudio)
3. Dataiku Academy: Intro to Machine Learning
4. Dataiku Academy: Machine Learning Basics
5. Dataiku Academy: Scoring Basics
6. Dataiku Academy: Time Series Basics
7. Dataiku Academy Use Cases: Predictive Maintenance, Churn Prediction, Web Logs Analysis, Network Optimization, Bike Sharing Usage Patterns

SEMESTER/YEAR : **IV SEM/ II YEAR**
COURSE CODE : **25MBAB005**
TITLE OF THE COURSE : **EDA USING PYTHON**
L: T: P: C : **3: 0: 2: 4**

Overview

The programming requirements of data science demand a very versatile yet flexible language which is simple to write the code but can handle highly complex mathematical processing. Python is most suited for such requirements as it has already established itself both as a language for general computing as well as scientific computing. Moreover, it is being continuously upgraded in form of new addition to its plethora of libraries aimed at different programming requirements.

Course prerequisites

- This is a hands-on practical course. Lab computers must have Anaconda (Python IDE) installation with Jupyter Notebook and working internet connection
- Students are not expected to have previous programming knowledge, but they must be willing to learn and practice beyond classroom sessions.

Course Objectives

1. To enable students, understand the features of Python language and different IDEs Like Jupyter Notebook and Spyder.
2. To enable students, understand the features and functions of Python Packages: Numpy, Pandas
3. To enable students to apply and leverage on Pandas and Numpy functionalities in Performing data handling, data manipulation and data management tasks.
4. To enable students to apply and leverage on Matplotlib Library to perform data visualizations and descriptive Data Analysis
5. To enable students to understand and apply the machine learning models using python language

Course Outcomes

By the end of this course, the students would

1. Exhibit the fair understanding of concepts, theories, models and principles of Python programming course, including its features and applications.

2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of Python programming in real-time problem solving by using NumPy for data handling and manipulation, and understanding the capabilities of the Pandas library.
3. Demonstrate the ability of analyzing real-time problems using the skills acquired from concepts, theories, models and principles of Python programming by examining NumPy and Pandas libraries for advanced data management tasks along with databases.
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of Python programming for descriptive statistics and data visualization tasks.
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of Python programming, including adapting and applying different machine learning algorithms.

Approach to Learning	● Lectures ● Active student participation and classroom exercises ● Mini Project ● Case Analysis collaboratively with students 'involvement ● Visual presentation
Assessment Strategy	Participants will be assessed on both conceptual understanding and business applications of Python Programming Language ● Case study discussions & solving mini projects ● Submission of assignments and individual presentation ● Written Exam

Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	**Introduction to the Data Analysis and Python World** Data Analysis Process; Quantitative and Qualitative Data Analysis Python Introduction; Python Interpreter; Python Distributions; Anaconda Environment; Jupyter Notebook; IDEs for Python; Built-in Python Libraries; Built-In Functions; Data types; Libraries and Modules. Programs on applications of Data Types methods and Bulletin functions.	8

Unit II	**Introduction To NumPy and Pandas Libraries** NUMPY: Creation of an Array; Operators; Functions; Indexing, Slicing, and Iterating; Shape Manipulation; Array Manipulation; Structured Arrays; Reading and Writing Array Data on Files. PANDAS: Introduction to Pandas Data Structures; Reindexing, Dropping, Arithmetic and Data Alignment; Operations Between Data Structures; Function Application and Mapping; Sorting and Ranking; ―Not a Number Data; Hierarchical Indexing and Levelling.	8
Unit III	**Data Handling with Pandas** Reading Data in CSV or Text Files; Reading and Writing HTML Files; Reading Data from XML; JSON Data; Pickle—Python Object Serialization; Reading and Writing Data on Microsoft Excel Files; Interacting with Databases; Data Preparation; Concatenating; Data Transformation; Discretization and Binning; Permutation; String Manipulation; Data Aggregation; Group Iteration; Advanced Data Aggregation.	8
Unit IV	**Descriptive Statistics and Data Visualization** Descriptive Statistics: Distribution Functions; Measures of Central tendencies; Measures of Variations; Measures of Shape. Visualizations: Visualization with Pandas Library; Visualization with Matplotlib Library.	8
Unit V	**Model Development and Evaluation** Hypothesis Testing; p-hacking; Correlation and Regression; Constructing and Implementing Linear Models; Descriptive; Supervised Machine Learning; Unsupervised Machine Learning; Reinforcement Learning; Statistics: Unified machine Learning Workflow; Exploratory Data Analysis on a Sample Dataset.	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	3	-	-	-	2
CO 2	3	-	-	-	2
CO 3	3	-	-	-	2
CO 4	3	-	-	2	2
CO 5	3	-	-	2	2

Action Based Component

- Mini Projects on Data collection, Data Cleaning, Data modification and Data Visualization followed by Data Analysis.

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	**2**	**3**	**4**	**5**	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester	20	X	X	X			

	Exam							
6	CBT	10	X	X	X			
7	Semester End Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Python Data Analytics with Pandas, Numpy and Matplotlib (Apress), Fabio Nelli 2018
2. Hands-On Exploratory Data Analysis with Python (Packt Publishing), Suresh Kumar Mukhiya and Usman Ahmed 2020

Reference books

1. Python for Marketing Research and Analytics (Springer International Publishing), Jason S. Schwarz, Chris Chapman, Elea McDonnell Feit (2020)

Readings & Case Analysis

1. Python Documentations from www.python.org

Suggested Courses

1. Python for Everybody Specialization - https://www.coursera.org/specializations/python
2. Crash Course on Python - https://www.coursera.org/learn/python-crash-course

3. Programming for Everybody - https://www.coursera.org/learn/python